

FORKLIFT THEORY STUDY



GUIDE

Legislative requirements

Hazards are all around us in a forklift operating environment. Later in this guide we will go into some of the common ones. But first let's define the word **'HAZARD'**:

- **Something that may cause harm or injury**

Whenever there are hazards, we also have risks. For example, you spill some water on a kitchen floor and the floor becomes wet. The wet floor is the 'hazard'. The risk is that you may slip over and hurt yourself. This word **'RISK'** is defined as:

- **What could happen to you when exposed to a hazard**

When you are working in a high risk environment where forklifts are, you have a legal "DUTY OF CARE" as a worker. But what does this involve?

- **Take care of your own safety**
- **Take care of others safety**

If you were to work in an unsafe manner whilst you hold a HRWL (high risk work licence) your safety regulator could take the following actions:



Safety Regulators:

NSW – SafeWork NSW
Victoria – WorkSafe Vic
Qld – Worksafe Qld
WA – Worksafe WA

- **Suspend your HRWL**
- **Cancel your HRWL**
- **Refuse to renew your HRWL**

If you need sources of information relating to safety you can check any of the following documents or references:

- | | |
|---|---|
| <ul style="list-style-type: none">• Operator's manual• Safety policies• Australian standards• Safe work procedures (SWP's) | <ul style="list-style-type: none">• Job Safety Analysis (JSA'S)• Safety Data Sheets (SDS)• Safe work load charts (SWL charts-racking)• Codes of practice |
|---|---|

It is up to our employer to **PROVIDE** us with everything we need to ensure we are safe and healthy at work. This includes:

- **PROVIDE** a safe workplace
- **PROVIDE** safe working equipment
- **PROVIDE** safe working systems
- **PROVIDE** safe facilities

So now you hold a High Risk Work Licence for a forklift (Class LF). If you get to work and are **unfamiliar with the forklift** you have been provided by your employer, they must provide:

- Training
- Supervision
- Information
- Instructions

Hazards – controls and planning

So we now know what a 'hazard' is. But before we start our work on the forklift we need to talk to other people in the workplace about these known hazards:

• Supervisors • Co-workers • Managers	• First aid officers • Fire wardens • Truck drivers
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But why do all these people need to be made aware of the hazards?

- To be aware of hazards in the workplace
- To identify hazards and controls

Let's now think about ourselves operating a forklift in the workplace. There are so many common **HAZARDS** in a forklift operating environment. Here's some examples:

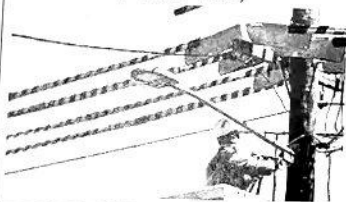
- | | |
|---|---|
| <ul style="list-style-type: none"> • Power lines • People • Other forklift traffic • Poor lighting • Poor ventilation • Poor weather conditions • Uneven surfaces • Wet surfaces • Blind corners • Obstructions | <ul style="list-style-type: none"> • Slopes or ramps • Overhead light fixtures • Damaged racking systems • Truck traffic • Loading docks • Excessive noise • Hot or cold environments • Surrounding structures • Dangerous goods |
|---|---|

Making powerlines more visible

Operating a forklift around power lines can be very risky. Especially if there is a lot of sun glare, making the power lines harder to see clearly. There are many different types of visual indicators used to help here:

Tiger Tails

(Black and yellow stripes wrapped around a power line)



Power line marker balls

(Bright orange balls evenly placed apart on the power line)



Warning signs



Operating forklifts around power lines

Each state or territory in Australia has their own government regulations when it comes to working near power lines

In **New South Wales**, for anyone who is not accredited, equipment operation may not be any closer than the following distances to electric/power lines

Electric/Power Line Type	Distance
Up to and including 132kV	3.0m
Above 132kV up to and including 330kV	6.0m
Above 330kV	8.0m

In **Victoria** the Framework for Undertaking Work Near Overhead and Underground Assets states that equipment must not be closer than the following distances to electric/power lines:

Electric/Power Line Type	Distance
Distribution lines up to and including 66kV (power poles)	6.4m (or 3.0m with a qualified spotter)
Transmission lines greater than 66kV (towers)	10m (or 8m with a qualified spotter)

The **Queensland** Electrical Safety Regulation breaks down the distances in detail. Exclusion zones are broken down not only by size of electric/power line but also by the competency level of the operator. This means that the requirements should be clarified with the electrical authority before work commences even if the distance appears to be outside the zones.

The following minimum distances are provided as guidance:

Electric/Power Line Type	Distance
Up to 132kV	3.0m
132kV up to 330kV	6.0m
330kV and above	8.0m

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In **Western Australia** we fall under Regulation Code 3.64. So in Western Australia these are the minimum safe distances when working near powerlines

Electric/Power Line Type	Distance
Live insulated powerlines 1000V or less	0.5m
Live uninsulated powerlines 1000V or less	1.0m
Live powerlines between 1000V and 33000V (insulated or not)	3.0m
Live powerlines exceeding 33000V	6.0m

Let's say you were operating the forklift near powerlines but you're unsure of the voltage. The best way to find out is to:

- **Call the local power company**

If you weren't paying attention and accidentally came into contact with the overhead powerlines whilst you were operating the forklift. There is a **6** step procedure that must be followed **IN ORDER:**

1. Warn people to stay away
2. Try to break away from the power lines
3. Stay in the forklift if safe. If not, exit the forklift and jump/shuffle 8m away with feet together
4. Do not touch any metal
5. Report to a supervisor
6. Check forklift before reuse

Hazards – controls and planning (continued)

During your day to day operations of the forklift you need to apply various **CONTROL MEASURES** to keep pedestrians or other vehicle traffic safe from your forklift operating area. These are some common examples:

- Warning signs
- Barriers
- Walkways
- Flashing lights
- Traffic mirrors



(These signs are bright yellow)



When you are operating the forklift at night or in dark areas always have:

- Adequate lighting





When the surface conditions become wet or slippery you must take **extra** caution when operating the forklift. Take certain precautions such as:

- **Slow down**
- **Brake slowly**
- **Turn slowly**
- **Avoid using ramps/slopes**

Types of forklifts

Forklifts come in different types. The type of forklift you use will depend on the task at hand.

Internal combustion (LPG, diesel & petrol) forklifts:
Not suitable in a space with restricted air flow because:

- **Fumes can kill you**

(Used outdoors)



The most suitable forklift in a confined area with limited air flow is:

- **An electric forklift**

(Used indoors)



All terrain forklifts:

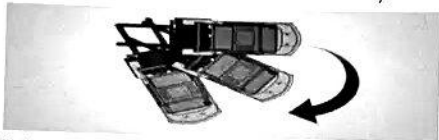
Only forklift suitable when the ground conditions are rough and boggy.

These forklifts can easily tip over due to having a higher centre of gravity.



Hazards – controls and planning (continued)

Most forklift have the steering wheels at the rear. This gives the operator the ability to take tighter and sharper turns. However, this creates a risk when the rear end is swinging round at about three times the speed of the forklift. This is known as 'REAR END SWING'. (Keep the front wheel close to where you are turning to control this movement)



This is dangerous because it:

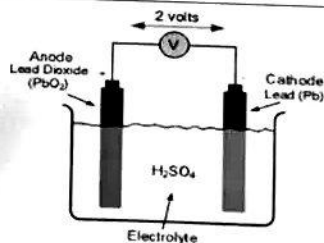
- Could hit nearby people or objects

Weather conditions can greatly affect our job as a forklift operator. It's always best to plan your forklift tasks around the weather forecasts as it will help plan:

- Helps to schedule the work tasks
- Helps plan for risk controls

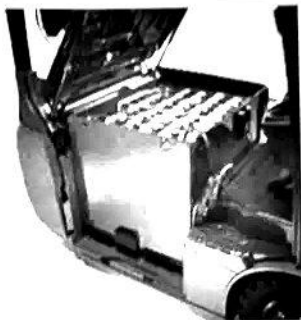
So we now know an electric forklift is most suitable where there is restricted air flow. However, the chemistry in these batteries is very dangerous when they on charge.

A wet cell battery consists of an acid solution (sulphuric acid and water). This is combined with a positive (anode) terminal and a negative (cathode) terminal, usually made up of lead and zinc plates.



When a forklift battery is on charge you are "boiling" the acid solution to generate more electricity into the battery. This process gives off a gas known as: "Hydrogen Sulphide".
THIS GAS IS EXTREMELY EXPLOSIVE!
Always keep the battery in a ventilated area, when on charge to:

- Release **EXPLOSIVE GASES** and prevent explosions



Refuelling the forklift

If the fuel powered forklift needs refuelling be sure to turn off the engine.
If the engine is still running whilst you are refuelling the forklift, there is a high chance of the:

- Fuel igniting
- Vapours igniting
- Gases igniting



Communications

There are dozens of ways to communicate with people in a workplace whilst you are operating the forklift such as:

- Written, Signage
- Verbally, Warning devices
- Hand signals, Two way Radio

Sometimes when you are operating the forklift co-workers need to communicate to you. The forklift can be noisy or you could be concentrating on what you are doing.

If a signal is given to you by a co-worker, that was unclear be sure to:

- Stop the forklift and ask to repeat the signal

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Planning your job

As a forklift operator you are going to be shifting loads around the workplace. The nature of a forklift is a "load shifting" piece of equipment. So ask yourself a series of questions like "What does my load weigh?" before you shift loads with your forklift.

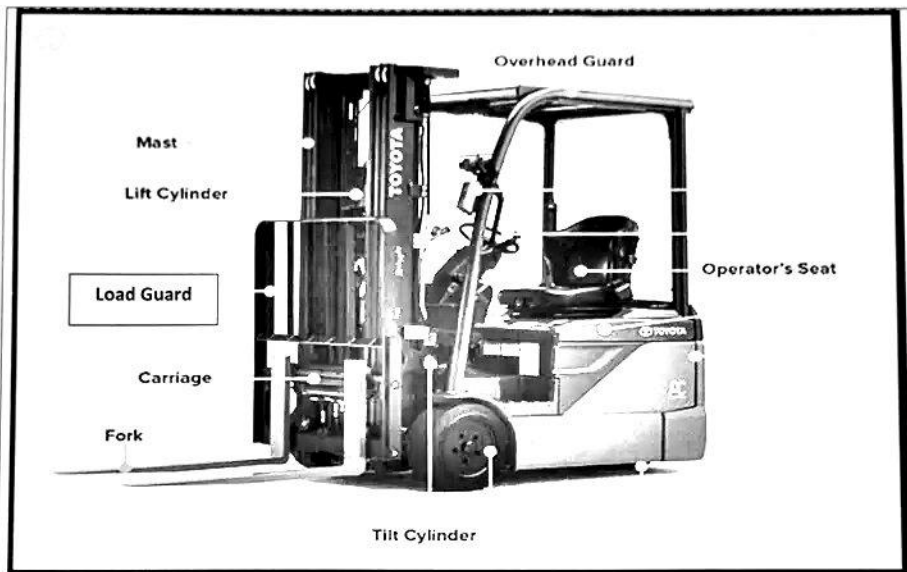
Things **OTHER THAN HAZARDS** need to be considered, before carrying out the shifting of various loads such as:

- **Load weight** (What is the load weight?)
- **Safest route to get to the load** (What is best path to get my load?)
- **Location of the load** (Where is my load located?)
- **Type of forklift** (Do I need a gas forklift? Or electric?)
- **Type of attachment** (What attachment do I need on my forks?)
- **Capacity of the forklift** (How much weight can my forklift lift?)
- **Communications** (safe and adequate)
- **Permits required**
- **Equipment required for the task**



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Pre-start inspection



Pre-starts need to be done **BEFORE YOU TURN ON** the forklift. You must check:

- | | |
|--|--|
| <ul style="list-style-type: none">• Log book and Operators/manufactures manual• Tyres & wheels• Fluid levels• Forks• Mast• Battery• LPG cylinder (gas forklift)• Data plate• Fork locking pins | <ul style="list-style-type: none">• Guards (Overhead & load)• Attachments• Hydraulic hoses• Chains• Hydraulic rams• Pulleys• Structural damage• Mirrors• Carriage• Seat• Seat belt |
|--|--|

Pre-start inspection (continued)

Forklifts fitted with solid or air filled tyres could have many defects when you are checking them over:

- Large pieces of rubber missing
- Flat tyres
- Unsecure wheels
- Underinflated tyres
- Uneven tyre tread
- Tyres worn down

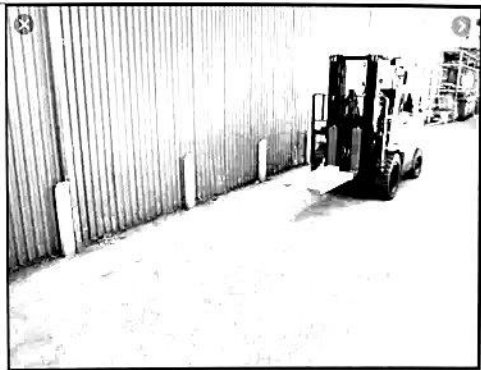


Attachments are likely to be used to shift odd shape loads such as rolls of carpet, stacks of tyres, 200L drums, engine blocks etc. If you are unsure about how to secure the attachment to your forklift, you can refer to the:

- **Manufactures/Operators manual** (for the attachment)
- **In the manufactures instructions that come with the attachment**

Attachments have a dramatic effect on the forklifts overall capacity. This photo is a "Carpet Spike". As you can see it is extremely long in length. This will greatly:

- **Reduce/Lower the capacity**
(Due to having a greater load centre distance)



Pre-start inspection (continued)

When you find a fault on the forklift such as the data plate being unreadable:

- Remove the key
- Tag "out of service" →
- Report to a supervisor



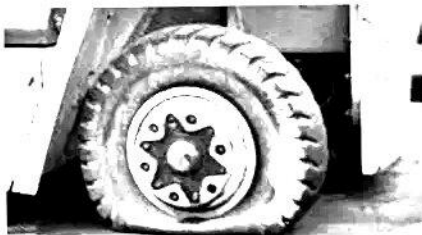
If you have to operate the forklift in different types of weather this can pose unique hazards:

- Wet/slippery conditions
- UV rays from the sun
- Cold/strong winds
- Fog limits visibility
- Extreme cold temperatures



Tyres on some forklifts are air-filled. It is important to ensure all tyres are at the same operating pressure:

- To keep the forklift AND load stable



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Start-up operational checks

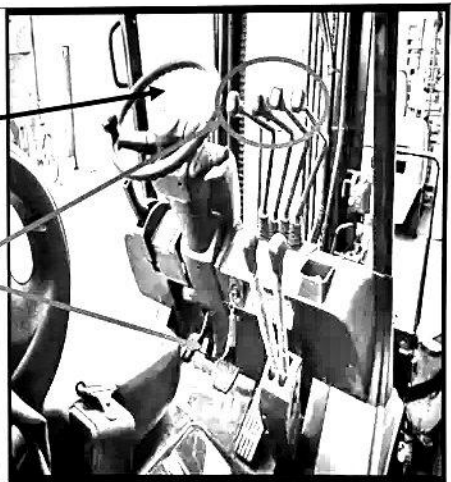
All forklifts are fitted with two guards, which have a function to protect:

- **Load Guard**
(Protects the mast and operator from the load)
- **Overhead Guard**
(Protects the operators head from falling objects)



Operational checks need to be done when you step into the forklift and **TURN ON** the ignition:

- **Steering**
- **Brakes**
- **Horn**
- **Lights**
- **Hydraulic levers**
- **Gauges**
- **Reverse beeper**



General operations

Before you insert the forks into the load you need to work out its weight. The following ways can be used to work out the weight of the load:

- Checking weighbridge documents
- Calculating the weight
- Checking the load for labels
- Weighing the load
- Checking the inventory system



A load can often obstruct your view due to it being too tall. You should always do these things when your view is obstructed:

- Travel in reverse
- Use a spotter
- Repack the load is safe to do so



While you are travelling with a load its movement needs to be **MONITORED**:

- To keep the forklift **AND** load stable

In very rare circumstances you are permitted to carry passengers on the forklift. Some sea container forklifts, like the one pictured are permitted to carry people. The only time it is allowed is:

- When it's designed to so (a 2 seater forklift rare but exist)



General operations (continued)

Whenever you travel with the load NEVER have it raised too high. This creates a high risk that the:

- Forklift could tip over



To prevent the forklift from tipping over always keep the load at a "safe height" when travelling:

- At axle height
OR
- As low as possible



Operating a forklift
a high risk to
or lower

Operating a forklift whilst carrying a load poses a high risk to people below the load. If you raise or lower the load over people, there is a:

- Risk of injury or death if load falls



Forklifts carrying a load can travel up or down ramps.

Always be sure to face the load:

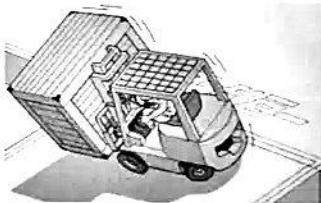
- Load must face Up the ramp



Travelling with a load on ramps or sloping surfaces is very dangerous if you are not facing the ramps squarely. If you turn a loaded forklift on a ramp:

It is unsafe to turn a loaded forklift on a ramp because

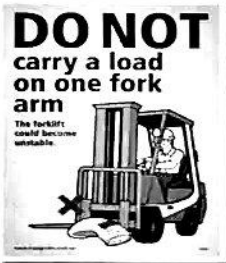
- You could lose the load
- You could tip the forklift



General operations (continued)

When you carry the load on only one fork arm this carries many risks and dangers. The most obvious ones are:

- Tipping the load
- Tipping the forklift
- Damaging the forklift



There are many **ACTIONS** that could cause the forklift to tip over **SIDEWAYS** when travelling:



NEVER TURN



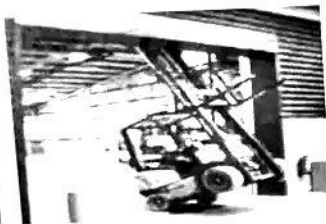
With An Elevated Load

- **TURNING** too fast
- **TURNING** on ramps
- **TURNING** with a load raised
- **Driving** on uneven surfaces
- Driving with an uneven/unbalanced load
- Carrying the load on one fork
- Uneven tyres
- Side shift not in centred

Actions to tip the forklift over forwards

There are many **ACTIONS** that could cause the forklift to tip over **FORWARDS** when travelling

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- DRIVING overloaded (overweight load)
- Braking severely/Heavy
- DRIVING with the load too high
- DRIVING too fast
- DRIVING with load facing forwards down the hill
- DRIVING over an uneven surface
- DRIVING with an uneven/unbalanced load
- Having too much forward tilt with the load raised
- Having the load too far forward on the forks (not against heels)

Attachments

There are a multitude of attachments available depending on the characteristics of the load. Here are some common ones at most workplaces:



- Drum CLAMP

(Used to pick up 200L drums)



- Paper Roll CLAMP

(Used to clamp large rolls of paper or tyres)



- Bale CLAMP

(Used to pick up crushed stacks of recycled cans or bales of hay)



- Fork extensions (Slippers)

(Used to pick up two loads at once or a load longer in length)

	• Jib (Used to swing odd shaped loads such as engine blocks or gearboxes)
	• Carpet Spikes (Used to pick up large rolls of carpet or artificial turf)
	• Personnel Work Platforms (Used to elevate people to perform maintenance or count stock)
	• Rotators (Used to rotate loads like a bin full of grapes into a wine hopper)

When an attachment is fitted to a forklift, the like effect to the capacity will be **"REDUCED"** so the lifting capacity will be lowered (**Reduced capacity**).

Also when using an attachment be sure to read and locate the **manufactures instructions (Operators Manual)** for the attachment which will advise you on correct use and fitting.

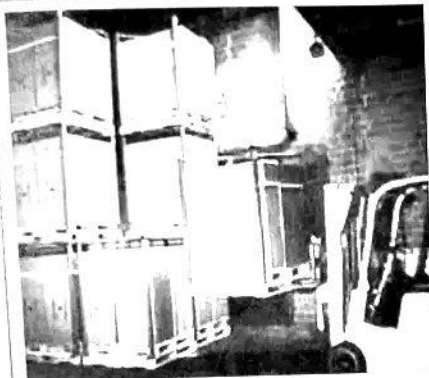
General operations (continued)

Most modern forklifts are equipped with a third hydraulic lever which makes the load guard shift left and right. This is commonly referred to as the:

"Side shift function"

This side shift function gives the operator the ability to square up a load or butt it up against an adjacent load next to it. But does carry some risks when using it. Always make sure to **"centralise"** the side shift when carrying a load:

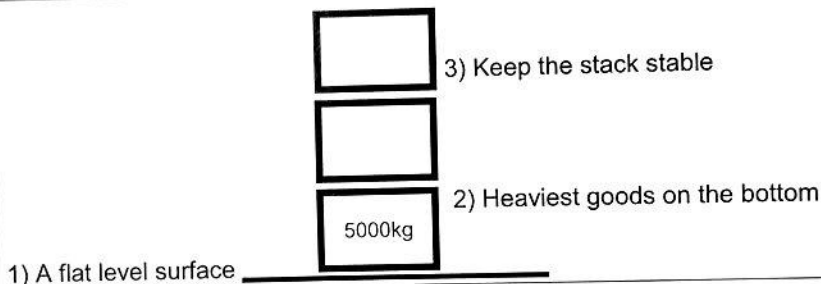
- **To keep the forklift AND load stable**



A common task that you may need to do as a forklift operator is stack loads on top of each other. Always plan for these **THREE** things when stacking the loads up on top of each other:

1. A flat level surface
2. Heaviest goods on the bottom
3. Keep the stack stable

The diagram below explains this process



General operations (continued)

If you were to turn around a corner too fast there is a high chance that your forklift could tip over sideways. It is very important to look out for your own safety if this is happening. You will find labels on the forklift looking like this:



If your forklift is tipping over sideways, you must always do the following to protect yourself from being crushed by the forklift:

- **Stay in the forklift (NEVER JUMP OUT!)**
- **Brace yourself**

- Lean away from tipping side

General operations (continued)

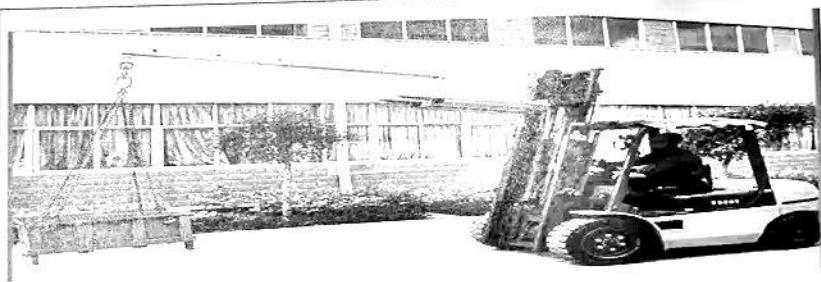
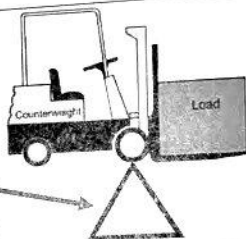
Most forklifts are equipped with a seat belt. The function of the seatbelt is:

- To stop you falling out in an accident



Every counter-balance forklift has a balance point. Its referred to as the "FULCRUM"
You can see on this diagram that it is indicated using a **triangle** under:

- The front tyres touching the ground



Jib attachments pose certain hazards and risks that need to be identified before and when operating the forklift with them attached. Because a jib acts as a "crane" for the forklift there can be a lot more movement of the load when you pick it up and travel with it. This can cause:

- A lower capacity
- A Swinging load

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If you had a catastrophic failure to your brakes, there are **TWO** things you should do:

If you had a catastrophic (major) failure to your brakes, steering or hydraulic systems during operations there are **TWO** things you should do:

- Stop forklift & Tag "out of service"
- Report to a supervisor



Loading Docks

Sometimes you need to unload goods by driving directly into a sea container. Trucks often back up into "Loading Docks" (see photo below)



A gap is then created between the back of the truck and the front of the loading dock. You must place either:

- Dock Plates
- OR
- Bridge Plates

Across this gap before shifting a load.

Take note of the data plates below and see the dimensions for the load centres DIM B and DIM C. These can be calculated simply by dividing the length (DIM B) and height (DIM C) by 2.

DIM B is the horizontal load centre

DIM C is the vertical load centre

For example if a load has a length of 1200mm and a height of 1500mm then the horizontal load centre is 600mm (DIM B) and the vertical load centre is 750mm (DIM C). Workings out below

Length is 1200mm, $1200 \div 2 = 600\text{mm}$ DIM B

Height of load $1500\text{mm} \div 2 = 750\text{mm}$ DIM C

This is an example of the forklifts:

- **Data plate**

It is where you will find the forklifts rated capacity.
(10000kg on this forklift)

Trained Operators and Mechanics Only

Read Operating Manual located on or near seat

Failure to follow operating, maintenance and maintenance instructions can cause serious injury or death.

CAPACITY WITH MAST VERTICAL OR MAST TILTED FORWARD AND EQUIPPED AS SHOWN

HYSTER Lift Truck Model: H12.00AM-12EC

Serial No: F019E01690B
Attachment: 2500 mm SS Ctrp + 2440 mm Forks

Truck Weight: 20650 kg
Forward/Back Tilt: 6/5.0 Degrees
Front Width: 2870 mm
Type: Front
Size: 12.00X20 Dual Pne
Pressure: 750 KPA

Load Height Dim A	MAXIMUM CAPACITY Mast Vertical	Load Centre		MAXIMUM CAPACITY Mast Tilted	Load Centre	
		Dim B	Dim C		Dim B	Dim C
3700 mm	10000 kg	600 mm	600 mm	10000 kg	600 mm	600 mm
3700 mm	10000 kg	1220 mm	1220 mm	10000 kg	1220 mm	1220 mm

Load centres

FORKLIFT TRUCK			
MODEL NO.	FL1101	SERIAL NO.	F1000140C098
FORK ARMS	MAST VERTICAL CAPACITY	MAST FORWARD CAPACITY	MAX HEIGHT OF LIFT
	2250 kg	1650 kg	3500 mm
			LOAD CENTRE DISTANCE
			600 mm

Other limitations and important information about the forklift and equipment can be found in the operator's manual and manufacturers documents.

Load Centre Distance

Load Height

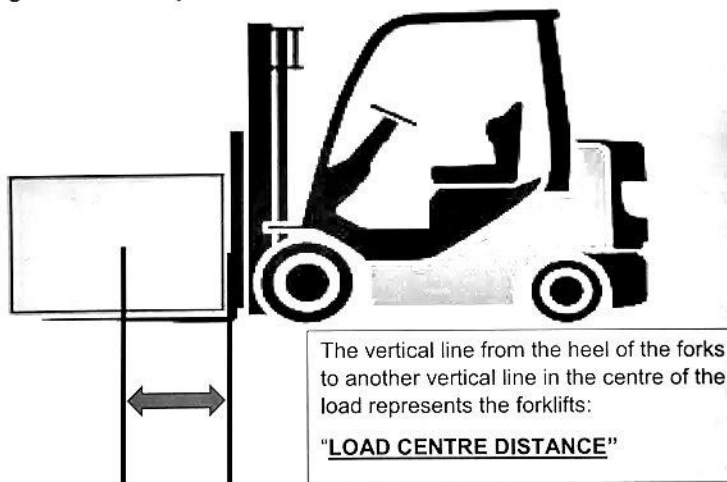
Maximum Load

The forklifts data plate above is rated at 2250kg, to a lift height of 3500mm with a load centre distance of 600mm (or less). What does this 'load centre distance' mean?

The manufacturer measures this distance:

- From the vertical face (heel) of the forks to the loads centre of gravity

This diagram will show you:



Load centres (continued)



This photo has a load that is **NOT** against the heel of the forks. The **FORKLIFT** will be affected in two ways:

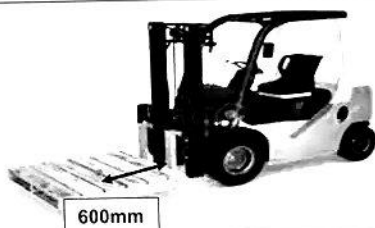
- A lower capacity
- OR
- The forklift could tip over

Most forklifts in Australia are rated at a load centre distance of:

- **600mm**

This comes from half the width of a standard pallet.

(1200mm x 1200mm - Length X Width)



Shut down

There are many places in a workplace that a forklift should **NOT** be parked in. Here are some examples:



- Near first aid stations
- Near emergency exits
- Near fire extinguishers
- On ramps
- On walkways

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There are several steps that need to be done when parking a forklift:

- Tilt forks forward and lower to the ground
- Neutral gear
- Hand brake on
- Remove the keys
- Turn off the LPG cylinder
- Place battery on charge



When the forklift has been finished with and you park up for the day always ensure you remove the key:

- To prevent unauthorised use



Calculations

Part of your job as a forklift operator is to do some basic maths calculations to work out the weight of a load. Let's do some examples:

A pallet of cement bags (see photo):

- Five bags to a layer
- Eight layers on the pallet
- Each bag is 25kg
- The pallet weighs 30kg

$5 \times 8 = 40$ bags (work out the total items first)

$40 \times 25 = 1000$ kg (work out the total item weight next)

$1000 + 30 = 1030$ kg (work out the grand total weight last)

Always show all your calculations



A pallet of drums (see photo):

- Six drums on the pallet
- Each drum weighs 150kg
- The pallet is 35kg

$6 \times 150 = 900$ kg (work out the total item weight first)

$900 + 35 = 935$ kg (work out the grand total weight last)

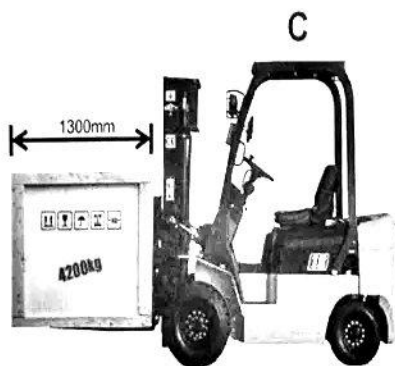
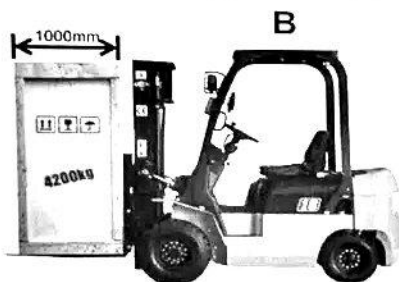
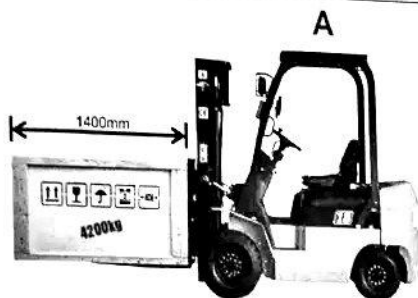
Always show all your calculations



Load centre distances

Let's briefly go back to load centre distances to work out how to find the load centre of any given load.

- The forklift below is rated at 4200kg at a 600mm load centre



To work out the load centre distance of each of the loads is a simple maths equation:

Load width (or length) divided by 2 = Load Centre Distance

Load A: $\frac{1400}{2} = 700\text{mm load centre}$

Load B: $\frac{1000}{2} = 500\text{mm load centre}$

Load C: $\frac{1300}{2} = 650\text{mm load centre}$

Then you check each calculated load centre against your forklifts load centre and see which one is within the specifications.

In this case Load B is the only one within this forklifts specifications as its load centre is 500mm.

Data plates

FORKLIFT DATA PLATE				(example only)									
MODEL	SERIAL NO	MAX HEIGHT	MAX FWD TILT	<u>Tyre</u> <u>Front</u> <u>Rear</u> Size: 200/50-10 Solid 15x4.5-8/3.0 Solid Tread width: 908mm Pressure: N/A									
J2.0 XYZ	ACLF-20	4200mm	5 DEGREES	TRUCK WEIGHT 5050 kg FUEL TYPE: LPG									
MAST VERTICAL				MAST FORWARD TILT 5 DEGREES									
ATTACHMENT TYPE	FORK HEIGHT mm	LOAD CENTRE mm	RATED CAPACITY kg	<table><tr><th>FORK HEIGHT mm</th><th>LOAD CENTRE mm</th><th>RATED CAPACITY kg</th></tr><tr><td>4200</td><td>600</td><td>2050</td></tr><tr><td>4200</td><td>1400</td><td>750</td></tr></table>	FORK HEIGHT mm	LOAD CENTRE mm	RATED CAPACITY kg	4200	600	2050	4200	1400	750
FORK HEIGHT mm	LOAD CENTRE mm	RATED CAPACITY kg											
4200	600	2050											
4200	1400	750											
1070mm Forks and Side shift	4200	600	2500										
3000mm Carpet Spike	4200	1800	1000										

In this example of a data plate the capacity of the forklift is given in **kg**. It has been split into **two** sections; **MAST VERTICAL AND MAST FORWARD TILT 5 DEGREES**

The capacity will change with a forward tilt mast compared to a vertical mast.

The **three** columns on each section give; **FORK HEIGHT (OVERALL LIFT HEIGHT), LOAD CENTRE (HEEL OF FORKS TO LOADS CENTRE) AND RATED CAPACITY (OVERALL WEIGHT THAT CAN BE LIFTED SAFELY)**

It is important that you understand how to read a data plate so you know what your forklift is capable of. Each manufacturer will use a similar layout to this example, but some manufacturers lay out their data plates differently. Always ask your supervisor if you are unsure how to read the forklift data plate

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Data plates (continued)



Lift Truck Model **H16.3 XL-EC4**

Year of Manufacture

S/O

Serial No. **D019E01561PS**

Attachment: **2500 mm SS Crrg + 2440 mm Forks**

Truck Weight **23270 kg**

Forward/Back Tilt **5/5 Degrees**

Tread Width **2870 mm**

Type **Front**

Rear

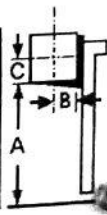
Size **12.00x20 Dual Pneu**

12.00x20 Pneu

Pressure **750 kPa**

750 kPa

Lead Height Dim. A	MAXIMUM CAPACITY Mast Vertical	Load Centre		MAXIMUM CAPACITY Mast Fwd Tilted	Load Centre	
		Dim. B	Dim. C		Dim. B	Dim. C
6200 (mm)	10000 kg	1200 (mm)	1200 (mm)	7010 kg	1200 (mm)	1200 (mm)
9910 (mm)	9000 kg	1200 (mm)	1200 (mm)	6120 kg	1200 (mm)	1200 (mm)



Let's have another look at a forklift data plate. This forklift is rated at 9000kg with a horizontal load centre distance of 1200mm (Dim. B) And a vertical load centre of 1200mm (DIM C) At a lift height of 9910mm.

To work out whether a load can be lifted by the forklift, you must look at **TWO** things.

- The load weight (measured in kg), AND
- The load centres of the load (measured in mm)

Let's look at some loads and work they are in the forklifts capacity. **(TURN TO PAGE 32)**

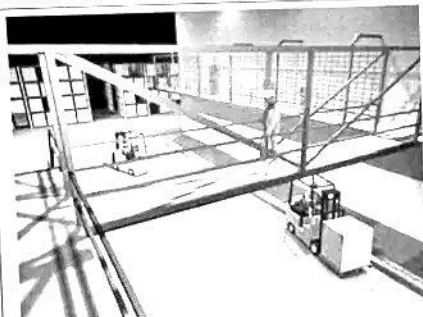
Load Information	Reason/Calculations	Yes or No?
<u>A pallet of bags</u> - The bags and the pallet weigh 2500kg - The pallet is 1180mm X 1180mm (Length X Width) - The pallet/load height is 1400mm	1) Weight and load centres need to be compared to the forklifts capacity. This is done by simply writing: $1180 + 2 = 590\text{mm}$ less than Dim B 1200mm $1400 + 2 = 700\text{mm}$ less than Dim C 1200mm Load weight 2500kg less than 10000KG (7500kg less) YES it can be lifted 2500kg IS LESS THAN 9000kg CAPACITY LOAD CENTRES ARE LESS THAN FORKLIFT RATING OF (1200mm)	Yes! Then you can tick the box "YES" to answer the second part of the question
<u>A loaded shipping container</u> - The 'tare' (weight of the empty container) weight of the container is 5000kg - The load contained inside the container is 7000kg - The shipping container size is 2.5m wide, 2.6m high and 2.5m long	1) Weight needs to be compared to the forklifts capacity. We don't know the total weight of the loaded sea container yet. So we need to start with a calculation. Weights 5000kg(Tare/empty) plus 7000kg(contents/load inside container) Calculation $5000\text{kg} + 7000\text{kg} = 12000\text{kg}$ Load centres 2.5m (2500mm) width divide by 2 2.6m (2600mm) Height dived by 2 Calculation $2.5\text{m} \div 2 = 1.25\text{m}$ (1250mm) over DIM B $2.6\text{m} \div 2 = 1.3\text{m}$ (1300mm) over DIM C We can clearly see this load is over the rated capacity and load centres	No! Then you can tick the box "NO" to answer the second part of the question

Other information (For Help with Our Assessments)

An operator needs to be aware that most forklifts are designed for hard flat surfaces such as hard bitumen or flat concrete surfaces. The following surfaces should be avoided at all times when operating any type of forklift:

- | | |
|--|---|
| <ul style="list-style-type: none">• Backfilled ground• Damaged cracked concrete• Trench covers or grates• Soft soils• Railway tracks | <ul style="list-style-type: none">• Potholes• Rough uneven surfaces (rough terrain trucks okay here)• Hard compacted soil• Damaged cracked bitumen• Sloping surfaces (High reach truck forklifts are not suitable due to being top heavy) |
|--|---|

Pedestrians are most at risk when it comes to the forklift operating area. There are many ways to separate them from the forklift operation area. Here are some common examples used in many workplaces:



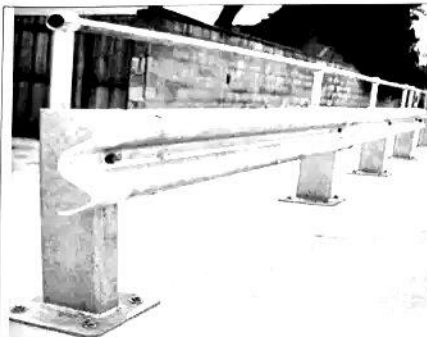
Overhead walkways and vehicle routes.

These are used to suspend people above the dangerous forklift operating area.



Designated walkways and parking areas.

Usually bright yellow stripes painted across the floor, with a "foot" symbol painted at intervals to indicate pedestrian walking area only.



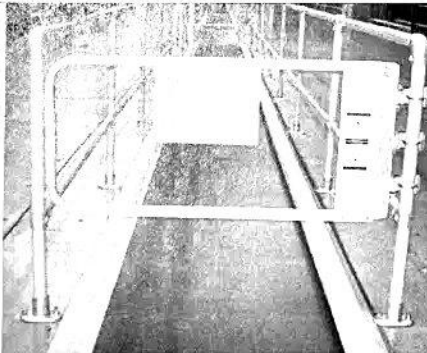
High impact barriers

Strong galvanised steel barriers, able to withstand a larger impact.



Pedestrian crossings

Bright yellow stripes painted like "Zebra Crossings" for people to walk over.



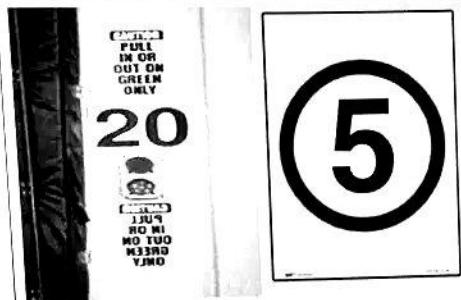
Gates and bollards (or guardrails)

The bollards are usually bright yellow so they stand out clearly. The gates have spring hinges so they close automatically behind the passing pedestrian.



Temporary barriers

Simple, but effective barriers bright yellow in colour, that can erected quickly and moved around if needed.



Flashing lights and signs – for example “Keep to a speed limit of 5km/h at all times”

Flashing lights are usually found on the back of loading docks to indicate to a forklift driver that a truck is now safe to unload or load.

Speed limit signs are white, red and black.

Notes

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